Garrett Comes

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Simulation & Scientific Software

Valurile — GPU Lattice-Boltzmann CFD Solver | *Rust, CUDA* **2024 – present**

- Multi-crate LBM solver built for memory-bandwidth-bound performance: Esoteric Pull in-place streaming that roughly halves memory traffic per timestep, static mesh refinement, Smagorinsky–Lilly LES, SRT/TRT/MRT collision operators.
- Interactive egui/GPU analysis frontend: airfoil and cylinder obstacles, physical-to-lattice nondimensionalization, velocity/density/pressure/vorticity fields, drag and lift coefficient histories.

YBCO Coil Quench Simulation — SCRAP senior design | *Python, FEniCSx, Gmsh, PETSc* **2025 – 2026**

- Coupled electromagnetic–thermal model of a high-temperature superconducting pancake coil: H-formulation EM with E–J power-law resistivity, per-turn tape meshing, staggered thermal coupling with radiation and cold-finger BCs, quench detection, and adaptive time-stepping; supports an AIAA SciTech 2027 extended abstract in preparation.

spacedb — Space-Object Database & Route Planner | *Rust, egui, wgpu* **2025 – 2026**

- Native GUI + CLI over the full public satellite/debris catalog: SGP4 and high-fidelity propagation with drag, SRP, and third-body effects, parallel collision screening, decay prediction, value scoring, and multi-target servicing-route optimization with an interactive wgpu 3D scene.

ai-cfd — GPU-Resident Solver for ML Dataset Generation | *Python, Triton* **2026**

- Staggered-grid incompressible flow solver written entirely in Triton kernels; adaptive timestep, Jacobi/CG/BiCGStab pressure solves, per-step divergence, residual, and drag/lift diagnostics, and traceable train/val/test dataset sweeps for AI-CFD surrogates.

fem_rust_ae — FEM Solver in Rust, live in the browser | *Rust, WASM, Bevy* **2025**

- From-scratch structural FEM with sparse solves via faer, an interactive 3D UI, and mesh-refinement convergence studies; compiled to WebAssembly and running live at garrettcomes.me/fem_rust_ae.

Thermodynamic Sampling Unit Research | *JAX, THRML, PyTorch* **2026**

- Joint block-Gibbs factor-graph decoding of a masked-diffusion LM on Extropic's TSU simulator, beating independent argmax by 7–15 pp; reproducible 59-landscape Potts coverage study with adversarial review gates on all headline claims, bit-for-bit reruns, and 9/9 regression tests passing.

Eddy-Current Braking System — Controls & Hardware | *Rust no_std, Arduino Due, NGSolve* **2025**

- End-to-end controls project: no_std Rust firmware with magnetic encoder feedback and feedforward-plus-PI speed control, a live egui tuning dashboard over UART telemetry plotting internal controller signals, and Python/NGSolve magnetic simulations of the braking physics.

Experience

Lockheed Martin Space — Propulsion Engineering Intern, Civil & Commercial Space **Summers 2022 – 2025**

Flight programs incl. Dragonfly and Mars Sample Return *Waterton Campus, Denver, CO*

- Designed a piston-tank propellant gauging system from scratch, hermetic internal magnets sensed by rad-hard Hall-effect sensors; built propellant slosh CFD models correlated against a test rig; contributed to a hydrazine-replacement green-propellant trade study.

Education

Georgia Institute of Technology **Jul 2022 – Aug 2026**

B.S. Aerospace Engineering, Concentration in Propulsion *Atlanta, GA*

- Coursework: Computational Fluid Dynamics, Finite Element Analysis, Jet & Rocket Propulsion, Aerothermodynamics, Control System Design.

Skills

Core: Rust, C++, Python, CUDA/Triton, JAX, PyTorch, FEM/LBM/FVM numerical methods, GPU memory-bandwidth optimization, wgpu/OpenGL

Tooling: Linux/NixOS, Git, MPI/PETSc/FEniCSx, eBPF, WASM, agentic AI development with Claude Code, MCP servers, and custom agents